

MANOHAR.G

Innovation | Consulting | Pre-sale

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Portfolio : https://manohargov.github.io/Portfolio

Professional Synopsis

- Over 23+ years of experience in Artificial Intelligence, Machine Learning, Advanced Analytics, and manufacturing across automotive, oil & gas, aerospace, medical, and robotics industries, driving enterprise AI, digital transformation, and product strategy initiatives.
- Led end-to-end pre-sales engagements, including client discovery, requirements analysis, solution consulting, proposal development, RFP/RFI responses, effort estimation, and business value mapping to accelerate AI adoption and revenue growth.
- Partnered with executive leadership and cross-functional teams across Marketing, Finance, Engineering, Product, Operations, and Compliance to define and execute strategic business and technology initiatives.
- Built, mentored, and led high-performing teams while fostering a culture of innovation, collaboration, and continuous learning.
- Established AI governance frameworks and Responsible AI practices, ensuring regulatory, risk, security, and compliance requirements.
- Proven track record of delivering enterprise-scale AI programs with a focus on operational excellence, KPI-driven execution, and measurable business outcomes.

Areas of Expertise

Strategy & Vision	Product & Innovation	Program & Delivery	AI Transformation	Python
Change Management	Governance	Statistics	Linux	GIT
PyTorch	Keras	PowerBi	MLOps	Scikit-learn
Numpy	Pandas	OpenCv	NLP	Azure
Six-sigma	LLM	MLflow	Autoware	ROS2
Ubuntu	CAD Automation	Paper replication	Manufacturing Technology	

Professional Experiences

Since-Feb'25 with **MSM (Remote)** **Program Manager-Ai**

- Managed end-to-end pre-sales activities — client discovery, requirement gathering, solution consulting, proposals, RFP/RFI responses, PoC planning, effort estimation, and business value mapping — to drive AI solution engagements and revenue growth.
- Led end-to-end delivery of Agentic AI programs across business domains, managing scope, timelines, budgets, and risks while establishing scalable Agile/SAFe delivery frameworks.
- Orchestrated implementation of AI agents, copilots, and autonomous workflow platforms, collaborating with architects, data scientists, and platform teams to define human-agent collaboration and escalation models.
- Embedded responsible AI practices — governance, explainability, and compliance — throughout agent deployment and optimization.
- Partnered with business leaders to translate requirements into AI-driven roadmaps, delivering executive reporting and value realization metrics while driving change management and adoption.
- Established enterprise AI governance covering data privacy, security, and model lifecycle management, ensuring alignment with enterprise architecture.
- Built reusable delivery accelerators and AI operational playbooks, defining KPIs and driving continuous improvement through analytics and feedback loops.

Projects:

E-Commerce -Agentic AI Platform

Designed a multi-agent AI system for e-commerce operations, orchestrating three specialized LLM-driven agents — Customer Requirement, Stock Clearance, and Transit Management — over an event-driven micro-services architecture. Automated customer intent capture, inventory clearance strategy, and shipment routing/exception handling, reducing manual coordination across sales, inventory, and logistics workflows.

Stack: Python, Ollama, DeepSeek-R1 14B, Sentence Transformers, all-MiniLM-L6-v2, GitHub Actions, Pytest

Gen3D -3D Model Generation

Built a Python tool that converts natural-language prompts into parametric CAD models. An LLM generates FreeCAD API code, which is validated, executed headlessly, and exported to STEP/STL. Added self-correcting error-recovery loops that feed failures back to the model. Cut routine part modeling from ~30 minutes to under a minute.

Stack: Python, FreeCAD API, Gpt-3 API

TSA-Tolerance Stack-Up Analysis
Built a Python tool that extracts dimensions and tolerances from engineering PDFs, performs stack-up analysis, and automatically generates reports—reducing manual engineering effort.
Stack: Python, GPT API, OCR, PDF Processing

CRA- Client and Revenue Analysis
Built a data-analysis solution that processed vendor data dumps, segmented potential clients, and ranked high-value opportunities based on project fit and revenue potential, enabling sales teams to target the most promising prospects.
Stack: Python, Pandas, SQL, Machine Learning, Power BI

Nov'13-Jan'25 with **Quest Global** , **Bangalore** **Senior Lead Engineer**

- Led end-to-end AI initiatives across the automotive, oil & gas, aerospace, medical and robotics sectors, driving strategic growth by delivering measurable business outcomes.
- Oversaw client relationships, business development, and P&L for complex AI and machine learning programs, ensuring project profitability and stakeholder satisfaction.
- Directed the full AI solution lifecycle, from conducting initial requirements analysis and developing proof-of-concepts to ensuring seamless deployment and operational success.
- Developed, tuned, and deployed complex AI/ML models, specializing in predictive analytics, forecasting, and computer vision for industrial applications.
- Architected and implemented scalable edge-to-cloud solutions, utilizing MLOps and LLMOps pipelines to ensure continuous delivery and robust performance.
- Led process optimization and efficiency enhancements using Agile and Six Sigma (DMAIC) methodologies, significantly improving supply chain and product development workflows.
- Monitored automation and model performance through metrics and reports, ensuring that quality, cost, and delivery objectives were consistently met.
- Directed and mentored a team of 10 ML engineers, overseeing project allocation, managing performance, and coordinating cross-functional, onshore/offshore development.

Algorithm worked & learned:
Task : Regression,Classification,Ensemble,Dimensionality Reduction,Clustering,Time Series, NLP, object detection & Classification, Multi-model, product based

Agentic/GenAI and Ecosystem:
Architecture : RAG based Systems,Model Fine-Tuning, Computer Vision
Models: OpenAi , Google Ai, HuggingFace Transformer
Supports: LangChain,LangGraph,LangSmith,FastAPI, Docker, Gradio, ChromaDB, FAISS.

Projects:
PPRS -Parts Proliferation Reduction System*: GenAI-Powered System for Cost Reduction Through Intelligent Prevention of Unnecessary New Part Creation.*

EPI -Engine Parts Identifier (Computer Vision) *: Engine component identification application powered by a fine-tuned EfficientNet model*

GCE -Geometric Cost Estimation (Computer Vision) *: Developed a YOLO-based object detection application for automated process cost estimation.*

OSO -Optimized Service Offering *: ML–driven optimization of supply chain costs, delivery timelines, and inventory storage efficiency.*

CPPD- Continuous Product and Process Development *: Architected a multi-module application integrating ML–driven capabilities across core functionalities.*

Nov'11-Sep'13 with **Quest Global**, **Chennai** **Lead Engineer**

Jun'10-Oct'11 with Armstrong International, Chennai **Product Design Engineer**

Jan'06-May'10 with, Larsen & Toubro-HED, Chennai **Design Engineer (Projects R &D)**

Jul'04-Dec'05 with E-core Engineers, Chennai **CAD Engineer**

Jun'03-May'04 with Swastika industries, Coimbatore **CNC Programmer | Supervisor**

Professional training & Achievements

- Autonomous Vehicle Systems Training by NITTTR Chennai Feb 2026 (Five Days)
- AI Agents Intensive Course by Google
- GenAI Intensive developer by Google
- Fundamentals of LLMs of The LLM Course By HuggingFace
- Fine-Tuning of LLMs Course By HuggingFace
- Certified Recruiter Training (Quest Global)

- Best Performer : Sep-2012, Aug-2013, Feb-2015, Oct-2016 (Caterpillar Inc.)
- Supplier Award of the Year: 2015 (Caterpillar Inc.)
- Best Performer of the Year: 2016-17 (Quest Global)
- Saved $2M - 2017-2018 by value Additions core change
- Marketed the product across peer divisions, driving adoption and generating $4M in new business revenue.
- On-the-Fly Award: Twice (Quest Global)

Academic Credentials

- **Advanced Certification in Data science & Artificial intelligence** - IIT-Madras -Intellipaat 2022 - 2023
- **Bachelor of Mechanical Engineering** – Apr 2014 on Satyabhama University, Chennai, India
- **Diploma in Mechanical Engineering** from AVC Polytechnic College, Mayiladuthurai in 2003.
- **Class X** Govt hr sec School, Thirunaipair in 1999.

Personal Dossier

Date of Birth : 12th May 1984
Language : Tamil, English
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Github : https://github.com/Manohargov